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TIE-BCPNN: Selective Conformal Risk Control with Audit-Trail and Counterfactual Diagnostics on Atomic-Posterior Bayesian Classifiers

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Abstract

We study conformal uncertainty estimation for a Bayesian Confidence Propagation Neural Network (BCPNN) used in a search-and-rescue (SAR) perception case study. The central difficulty is that the adaptive-prediction-set (APS) nonconformity score is highly discrete: approximately 0.87 of the calibration mass sits on the single value $s = 1$. This point mass, or *atom*, makes the split-conformal quantile degenerate and collapses the predictor to the trivial K-class set throughout the operator-relevant range $\alpha \in [0.05, 0.20]$.

We use this failure mode to evaluate TIE-BCPNN, a Trustworthy, Interpretable, Evidential extension of the BCPNN backbone. First, we document the at-$s = 1$ atom and show analytically why it causes split-conformal collapse. Second, we compare five atom-breaking variants: randomized APS, conformal temperature scaling (ConfTS), a structural Dirichlet floor, online adaptive conformal inference (ACI), and a TIE-modulated conformity score s_{TIE}. These variants recover informative marginal coverage, but they do not make conformal set size a reliable per-tile triage signal: across the canonical-ledger variants, AUROC[$|\mathcal{C}|$] remains in $[0.32, 0.50]$.

We therefore evaluate a selective-risk alternative based on a composite reject head using evidential uncertainty, maximum predictive probability, and conformal set size. Under an explicit bound$\times$head-refit ablation, SCRC-T achieves exact marginal validity at risk target $\alpha^\star = 0.32$ (nominal selective accuracy $1 - \alpha^\star = 0.68$), with empirical risk 0.247 (Wilson 95% confidence interval $[0.232, 0.262]$). To support auditability, we add two interpretability artifacts: an exact four-layer log-odds audit ledger with pool-wide residual 1.71×10^{-7} on $n = 11\,413$, and a per-hypercolumn simplex counterfactual perturbation diagnostic with validity 0.83 on $n = 100$ (Wilson 95% CI $[0.745, 0.891]$).

Finally, a no-fixed-point lemma explains why ACI cannot track $\alpha^\star$ when $\alpha^\star < m$ on atom-bearing scores; in our case the recursion empirically settles near $\alpha_\infty \approx 0.81$.

Keywords: conformal prediction; selective conformal risk control; adaptive conformal inference; evidential learning; counterfactual diagnostics; atomic-posterior Bayesian classifiers; BCPNN

1. Introduction

Picture a ground robot working its way through the rubble of a collapsed building. Its camera feed is divided into small image patches, which we will call *tiles*, and a classifier labels each tile with one of a handful of terrain categories. A remote operator uses those labels to decide where the robot can safely step, which parts of the scene to search next, and which parts to inspect personally. The two ways of being wrong are not symmetric. A tile that is confidently mislabelled can send a robot into a void or cause a region to be written off unsearched; a tile the system flags as doubtful costs only a few seconds of human attention. What the operator needs is therefore not merely a label, but a defensible statement of how far that label can be trusted — and, if something later goes wrong, a record of why the system decided as it did.

Conformal prediction is an attractive answer to the first half of that need. Instead of a single label it returns a *set* of labels, together with a finite-sample guarantee that the true label lies in the set at least a $1 - \alpha$ fraction of the time. The guarantee holds under exchangeability alone and does not require the underlying classifier to be well specified. Adaptive prediction sets (APS) (Romano et al., 2020) go one step further and aim to make the *size* of the set informative: an easy tile should receive a one-element set, an ambiguous tile a larger one. Read that way, set size is exactly the triage signal a search-and-rescue (SAR) operator wants — a cheap, label-free number that says "a human should look at this tile".

This paper reports what happened when we applied that recipe to a deployed SAR perception pipeline and it did not work. The backbone is the BCPNN-based sim-to-real terrain classifier of Cruz Ulloa et al. (2025). BCPNN denotes a Bayesian Confidence Propagation Neural Network: a Bayesian–Hebbian classifier whose posterior log-odds structure makes it unusually amenable to audit and explanation. On this backbone, split-conformal APS returned the same answer on essentially every tile — all $K = 5$ classes. Coverage was perfect and the output carried no class-level information whatsoever.

What goes wrong: an atom in the score distribution. The cause is a property of the score distribution that is simple to state and easy to overlook. Split conformal prediction

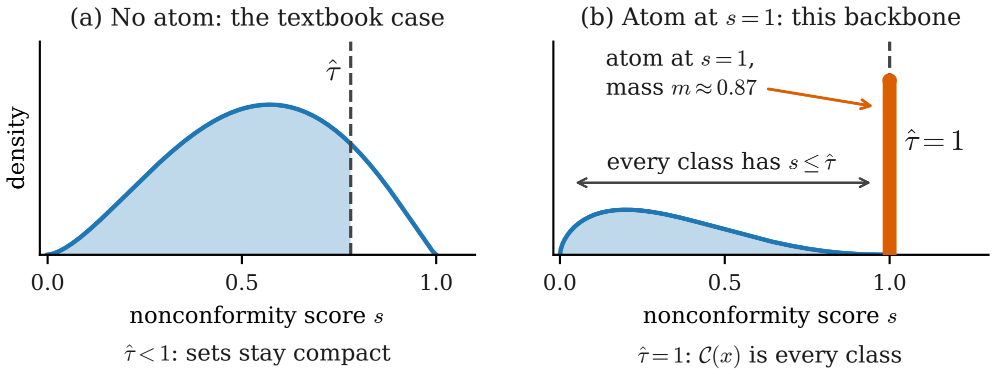


Figure 1: Why an atom collapses split conformal prediction (*schematic*; the only quantity taken from the experiments is the atom mass $m \approx 0.87$). *(a)* With a continuous score distribution the calibration quantile $\hat{\tau}$ lands strictly below the maximum, and the shaded $1 - \alpha$ of the mass falls to its left, separating classes inside the prediction set from those outside. *(b)* When a point mass — an *atom* — sits at the maximal score $s = 1$ and holds more mass than the target miscoverage α, leaving only $1 - m$ spread over $[0, 1)$, the quantile is pushed into the atom itself and $\hat{\tau} = 1$. Every class then satisfies $s \leq \hat{\tau}$ and the set degenerates to the full label space: coverage is preserved, information is not. Measured atom masses are in Figure 2.

turns a nonconformity score into a decision threshold by taking an empirical quantile of the calibration scores. That step tacitly assumes the scores are spread out, so that changing the quantile level changes the threshold. Here the assumption fails. The trained classifier is so close to deterministic that approximately 0.87 of the calibration tiles land on the same maximal APS score, $s = 1$, to within 10^{-7}. A distribution that places positive mass on a single point, rather than spreading its mass continuously, is said to have an *atom* there. The score distribution here is precisely such a mixture: a continuous part on $[0, 1)$ carrying about 13% of the mass, plus an atom of mass $m \approx 0.87$ at $s = 1$. Because the atom is so heavy, every quantile in the operator-relevant range $\alpha \in [0.05, 0.20]$ falls *inside* it and the fitted threshold is $\hat{\tau} = 1$; the inclusion rule "keep every class whose score is at most $\hat{\tau}$" then keeps every class. Figure 1 contrasts this with the textbook case.

Two questions follow, and their answers are the substance of this paper. The first is whether the atom can be removed without abandoning the backbone. It largely can: conformal temperature scaling (ConfTS), a structural Dirichlet floor, and online adaptive conformal inference (ACI) each restore marginal coverage close to target, with mean set size falling from the degenerate 5.00 to the range 3.50–4.03. Randomized APS, which only adds tie-breaking noise, does not escape the atom at all.

The second question is whether removing the atom restores what the operator actually wanted — that a large set means "check this tile". It does not. Across the variants measurable on a common per-tile ledger, the AUROC of set size as a per-tile error score stays within $[0.32, 0.50]$: uninformative at best, and at $\alpha = 0.10$ pointing the wrong way, with larger sets mildly *anti*-correlated with errors. Marginal validity and per-tile triage are different properties, and on this backbone only the first survives the repair. APS is designed to produce adaptive sets, not a calibrated per-tile error score, and this case study shows how far apart the two can drift.

That negative result reframes the rest of the paper. If set size cannot be the shield, something else must be, and it must carry its own guarantee. We therefore evaluate TIE-BCPNN (Trustworthy, Interpretable, Evidential BCPNN), an extension of the BCPNN backbone of Lansner and Ekeberg (1989); Lansner (2009); Ravichandran et al. (2025) combining recalibration, an evidential uncertainty head, conformal scoring, a selective reject head evaluated under explicit risk bounds, log-odds audit trails, and counterfactual perturbation diagnostics. The goal is not only to recover valid uncertainty estimates, but to determine which uncertainty signal is operationally useful for SAR triage.

The paper makes four contributions.

1. **Diagnosis of conformal collapse (§2).** We document the atom at $s = 1$ on the canonical BCPNN signals ledger, $m \approx 0.87$ of calibration scores satisfying $|s - 1| \leq 10^{-7}$, and show that the empirical split-conformal quantile $\hat{\tau}$ falls inside it for every $\alpha \in [0.05, 0.20]$.

2. **Atom-breaking study (§5.5).** We compare five ways to break or bypass the saturated APS score: randomized APS, conformal temperature scaling (ConfTS), a structural Dirichlet floor, online Adaptive Conformal Inference (ACI), and a TIE-modulated conformity score s_{TIE}. All restore informative marginal coverage; none makes conformal set size a reliable per-tile triage signal, with AUROC$[|\mathcal{C}(x)|]$ remaining in $[0.32, 0.50]$ across the canonical-ledger variants. We also prove Lemma 1, which explains why ACI cannot track a target $\alpha^{\star}$ below the atom mass m; empirically the recursion settles near $\alpha_{\infty} \approx 0.81$.

3. **Selective-risk alternative (§3.3, §5.3).** Because conformal set size is not a useful triage signal in this setting, we evaluate a composite acceptance rule based on evidential uncertainty, maximum predictive probability, and conformal set size. The rule is Pareto-tuned and evaluated under three selective-risk bounds: Beta-UCB, implemented as a Clopper–Pearson holdout PAC bound (Hoeffding, 1963; Clopper and Pearson, 1934), and the SCRC-I and SCRC-T bounds of Xu et al. (2025). We report an explicit bound×head-refit ablation, rather than comparing all bounds on a single fixed reject head.

4. **Interpretability artifacts (§3.1, §3.2, §5.1, §5.2).** We add two audit mechanisms tailored to the BCPNN representation: an exact four-layer log-odds audit ledger with a digamma-based Bayesian-shrinkage closure, achieving pool-wide residual 1.71×10^{-7} on $n = 11\,413$; and a per-hypercolumn simplex quadratic program (QP) computing the minimum-ℓ_2 hidden-activation perturbation that flips the argmax to a target class, which attains validity 0.83 on $n = 100$ real winner-take-all (WTA) saturated activations. We interpret the QP as a hidden-space diagnostic, not yet as an actionable input-space intervention.

Together these form a conformal, selective-risk, and audit-ready uncertainty pipeline for SAR sim-to-real perception, and the results are relevant to fairness, accountability, and transparency disclosure while keeping explicit which guarantees are marginal, which are selective-risk, and which artifacts are diagnostics rather than guarantees (§6). Our empirical claims are deliberately confined to one backbone and one corpus: this is a case study, not a survey. What does travel beyond it is the diagnosis and the test for it, and §6.1 gives both as a short recipe.

Roadmap. §2 fixes terminology, describes the data, and documents the atom empirically; §3 presents the four mechanisms of the trust layer; §4 contains the paper's one formal result, a no-fixed-point lemma for online adaptation under atom-bearing scores; §5 reports the empirical study; and §6 sets out what the evidence does and does not support. Readers wanting the short version can read §2.2 for the failure and §5.5 for the negative result that follows from repairing it.

2. Evaluation setup and atom-at-1 diagnosis

This section fixes the evaluation setup and the main failure mode studied in the paper. We first give the reader a glossary and a one-paragraph description of the data, then show why the APS conformal score degenerates on this backbone, and finally state the split protocol that determines which claims are formal and which are empirical.

2.1. Terminology, data, and components at a glance

This paper sits at the meeting point of three literatures — conformal prediction, evidential deep learning, and Bayesian–Hebbian (BCPNN) modelling — and few readers will be fluent in all three. Table 1 therefore gathers in one place the vocabulary and the components that recur later. Every entry is defined properly where it is first used; the table exists so that a reader who meets *hypercolumn* or *canonical ledger* in §5 need not hunt for it.

What the data look like. The corpus is the SAR terrain dataset of Cruz Ulloa et al. (2025). Images of terrain are divided into small patches, and each patch — a *tile* — is the unit that is classified, calibrated, and audited; every per-tile quantity in this paper is indexed by one such patch. The label space has $K = 5$ categories, of which class 0 is background and the remaining four are the terrain categories a SAR operator cares about. Two corpora are used: a larger simulation corpus of photorealistic IsaacSim renders, used to train the backbone and fit the calibration and evidential heads; and a smaller corpus of photographs of physical terrain, used to evaluate the trust mechanisms under the sim-to-real shift a deployed robot would face. The gap between the two is why the paper is careful throughout about which guarantee attaches to which split (§2.4). Background tiles carry no triage decision and are filtered out before the trust-mechanism experiments; the classifier itself keeps its original five-way output, which is why a fully uninformative conformal set still has size 5.

2.2. The atom-at-1 diagnosis

We now put the informal account of §1 on a measurable footing. Write F for the distribution of the APS-style nonconformity score $s \in [0, 1]$ on the calibration pool. We say that F has

Table 1: Reader's guide. Upper block: recurring terminology. Lower block: the seven TIE-BCPNN components, with the section defining each; these are the signals the experiments of §5 consume.

Term	Meaning
Tile	One image patch: the unit of classification and of every per-tile diagnostic.
$K = 5$	Size of the terrain ontology. Class 0 is background and is filtered from the trust-mechanism pool; a fully uninformative conformal set therefore still has size 5.
Hypercolumn, minicolumn	BCPNN hidden structure: H hypercolumns of M minicolumns each, whose activations are non-negative and sum to one within a hypercolumn (i.e. lie on a simplex).
WTA saturation	Winner-take-all regime in which nearly all of a hypercolumn's mass sits on a single minicolumn (Tully, 2016).
Atom, mass m	A point mass in the score distribution: one value carrying positive probability while the remaining mass is spread continuously. Here $\Pr[s = 1] = m \approx 0.87$ (§2.2).
$s(x, y)$, $\mathcal{C}_\alpha(x)$	The APS cumulative-probability nonconformity score (Romano et al., 2020), valued in $[0, 1]$, and the induced set $\{y : s(x, y) \le \hat{\tau}\}$ of size $\lvert\mathcal{C}\rvert$.
Selective risk	Error rate among the tiles a rule *accepts*, for a rule that is allowed to abstain.
SAC@$a^\star$	Accepted coverage of a rule whose selective accuracy is at least $a^\star$ (§2.4).
Canonical ledger	The fixed per-tile record of logits, calibrated probabilities, evidential quantities, conformal scores, and audit terms, computed once under §2.4 and reused by every experiment.
TIE-BCPNN components	
C1 Edge posteriors	Bayesian–Hebbian posterior log-odds $w_{ij} = \psi(C_{ij}) - \psi(C_i) - \psi(C_j) + \psi(C)$, with variance σ^2_{ij} (Lansner, 2009; Sandberg et al., 2002); see §3.1.
C2 Temperature scaling	A single global $\hat{T} \approx 12.0121$ fitted on $\mathcal{D}_{\text{cal-sim}}$ by NLL minimisation with L-BFGS-B (Guo et al., 2017).
C3 Evidential Dirichlet head	Converts the calibrated probability q into a Dirichlet with concentration $\alpha_y(x) \ge 1$ (§3). The floor is a *structural atom-break*: no class can take all predictive mass (Sensoy et al., 2018; Charpentier et al., 2020).
C4 Conformal layer	Four score families: randomized APS (Romano et al., 2020), ConfTS (Xi et al., 2025), online ACI (Gibbs and Candès, 2021), and s_{TIE} (§3.4.1).
C5 Audit ledger	Four-layer log-odds decomposition of a decision (§3.1).
C6 Reject head	Composite accept/abstain rule on $(U, \max_k p_{\text{dir},k}, \lvert\mathcal{C}\rvert)$, evaluated under the Beta-UCB, SCRC-I, and SCRC-T bounds (§3.3).
C7 Counterfactual QP	Minimum-ℓ_2 hidden-activation perturbation that flips the predicted class (§3.2).

an *atom* of mass m at $s = 1$ when

$$m \;=\; \Pr[s = 1] \;>\; 0,$$

that is, when a single score value carries positive probability instead of the mass being spread continuously over $[0, 1]$. Empirically we measure the atom at finite resolution, as the fraction of calibration scores with $|s - 1| \le 10^{-7}$. On this backbone that fraction is $m \approx 0.87$, and 65% of calibration scores are bit-exact at $s = 1$. The atom is not a numerical accident: it follows from the near-deterministic posterior geometry of the trained classifier, which we examine below.

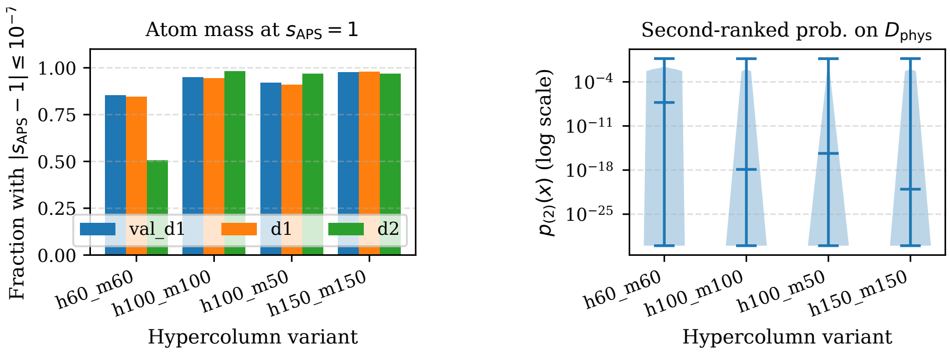


Figure 2: Atom-at-1 diagnosis on BCPNN APS-style scores. *Left:* fraction of APS-style nonconformity scores satisfying $|s_{\mathrm{APS}} - 1| \leq 10^{-7}$ across four hypercolumn variants and the available calibration/evaluation ledgers (shown in the legend). The atom mass is at least 0.83 in the physical-photo configurations used for the canonical analysis and reaches 0.87 on the canonical head. *Right:* second-ranked class probability $p_{(2)}(x)$ on the full post-class-0 $\mathcal{D}_{\mathrm{phys}}$ pool, shown on a logarithmic scale. The bimodality indicates two regimes: WTA-saturated tiles, where the runner-up class remains within about 10^{-3} of the winner, and near-deterministic non-saturated tiles, where the runner-up probability is essentially zero.

The consequence is immediate. For randomized APS (Romano et al., 2020), split conformal prediction forms a threshold $\hat{\tau}$ as an empirical quantile of the calibration scores and returns

$$\mathcal{C}_\alpha(x) = \{y : s(x, y) \leq \hat{\tau}\}.$$

Whenever $\alpha < m$, the $(1 - \alpha)$-quantile of F lies inside the atom, so $\hat{\tau} = 1$ and every class satisfies the inclusion test. The set is then the full ontology,

$$\mathcal{C}_\alpha(x) = \{1, \ldots, K\},$$

for every operator-relevant $\alpha \in [0.05, 0.20]$, since all of these are far below $m \approx 0.87$. The predictor preserves marginal coverage only by returning the uninformative all-classes set, even on tiles where the BCPNN has abundant evidence. This is the mechanism sketched in Figure 1(b). The argument uses no property specific to BCPNN, a point §6.1 returns to.

Figure 2 shows the measured version and why the score distribution takes this shape. The left panel confirms the atom mass across hypercolumn variants and ledgers; the right panel explains its origin. The second-ranked class probability $p_{(2)}(x)$ is bimodal: under winner-take-all (WTA) saturation the runner-up sits only about 10^{-3} below the winner, while otherwise the classifier is almost deterministic and the runner-up is effectively zero. Either way the mass sits at an extreme, leaving little intermediate probability for APS to work with. A score designed to interpolate between "obvious" and "ambiguous" has almost no intermediate regime to interpolate over.

2.3. Why atom-breaking restores coverage but not triage

Two mechanisms break the atom, in that limited sense, without changing the BCPNN backbone. ConfTS retunes the logits with $\hat{T} = 12.0121$, reducing the at-1 fraction from 0.87 to approximately 0.08; the Dirichlet head breaks it structurally through the constraint $\alpha_y(x) \geq 1$, giving marginal coverage in $[0.94, 0.96]$ and mean set size below the ACI baseline of 4.22 at $\alpha = 0.05$.

These repairs solve only the marginal-coverage problem. In both cases AUROC$[|\mathcal{C}|]$ remains close to 0.5, so set size does not reliably separate correct from incorrect predictions on individual tiles. The distinction matters for the rest of the paper: marginal validity answers "is the set valid on average?", whereas SAR triage asks "does *this* tile need escalation?" We make the claim rigorous in §5.5 with a five-variant study, and analyze the online ACI case in §4.

2.4. Dataset and split protocol

Throughout the selective-risk experiments, we write $\alpha^\star$ for the target miscoverage or selective-risk ceiling, so that the corresponding nominal selective accuracy is $1 - \alpha^\star$. We write $a^\star$ for a target selective accuracy. Under 0/1 loss these quantities are complementary, $a^\star = 1 - \alpha^\star$, but we keep both symbols because $\alpha^\star$ appears in the conformal-risk bounds whereas SAC@$a^\star$ reports accepted coverage at an accuracy gate.

We follow the BCPNN backbone, calibration head, and conformal split of Cruz Ulloa et al. (2025), writing $\mathcal{D}_{\text{phys}}$ for the physical-photo corpus described in §2.1. The split pipeline is

$$D_{\text{train-sim}} \to D_{\text{cal-sim}} \to \underbrace{D_{\text{phys-stage1}} \to D_{\text{phys-stage2}} \to D_{\text{phys-eval}}}_{50/25/25 \text{ split of class-0-filtered } \mathcal{D}_{\text{phys}},\ \text{seed } \texttt{20260425}},$$

with raw pool sizes $|\mathcal{D}_{\text{cal-sim}}| = 22\,880$ and $|\mathcal{D}_{\text{phys}}| = 15\,840$. After removing rows whose true label is background class 0, the trust-mechanism pool contains $n = 11\,413$ patches, split into $n_{\text{stage1}} = 2853$, $n_{\text{stage2}} = 2853$, and $n_{\text{eval}} = 5707$. Only evaluation rows with true label 0 are filtered; the classifier and all conformal predictors keep the original $K = 5$ ontology, so a full conformal set still has size 5 on this pool.

Table 2: Split protocol and role of each dataset. The formal risk-control claims are attached only to the physical-photo split assumptions stated here. They do not constitute a formal sim-to-real guarantee from the IsaacSim distribution to the physical-photo distribution. Such a claim would require an explicit distribution-shift correction, for example weighted conformal prediction under a covariate-shift model (Tibshirani et al., 2019), which this paper does not deploy.

Name	Source	n	Labels	Role
$D_{\text{train-sim}}$	IsaacSim	—	yes	Backbone fit
$D_{\text{cal-sim}}$	IsaacSim	22 880	yes	$\hat{T}$ and Dirichlet-head fit
$D_{\text{phys-stage1}}$	physical photo	2853	yes	Reject-head refit
$D_{\text{phys-stage2}}$	physical photo	2853	yes	Holdout risk calibration / Beta-UCB
$D_{\text{phys-eval}}$	physical photo	5707	yes	Coverage, AUROC, and size diagnostics

The 50/25/25 physical-photo partition is generated uniformly at random with fixed seed `20260425`. The SCRC-I/T evaluations rely on the exchangeability assumptions of their respective protocols (§3.3); Beta-UCB is treated as a Clopper–Pearson holdout bound on $D_{\text{phys-stage2}}$. Results on $D_{\text{phys-eval}}$ are reported as held-out empirical diagnostics of the selected mechanism. They are not used to claim that ordinary conformal prediction transfers validly across the sim-to-real boundary.

All experimental claims in §5 use this split protocol, and we claim only the SAR-realistic operating gate SAC@0.70 $\geq$ 0.30. Here SAC@$a^\star$ denotes selective accepted coverage at target selective accuracy $a^\star$: the fraction of the evaluation pool accepted by a rule whose selective accuracy meets or exceeds $a^\star$, equivalently the accepted coverage under selective-risk ceiling $1-a^\star$. Thus SAC@0.70 is an accepted-coverage value at a 70% selective-accuracy gate, not a risk-threshold statement. We do not claim SAC@0.85, which corresponds to a stricter target and is reported only when explicitly marked as a non-claimed reference point.

3. Methods

The diagnosis of §2 leaves the operator without a usable per-tile signal, so this section builds the trust layer that replaces it. Four mechanisms are described: two that explain a decision after the fact (an audit ledger and a counterfactual diagnostic), one that decides whether to accept a decision at all (the reject head), and one that adapts the conformal threshold online (the ACI shield). Only the third carries a risk guarantee; the first two are diagnostics, and we are explicit about that distinction in §6.

We treat the trained BCPNN backbone as a fixed feature extractor for the trust-layer evaluation. For each tile x, the backbone produces hidden posterior means $h(x) \in \mathbb{R}^{H\times M}$, organized into H hypercolumns with M minicolumns each, and class logits $\ell(x) \in \mathbb{R}^K$. The trust layer then applies the temperature $\hat{T}$ from C2 and the evidential Dirichlet head from C3. Let $q_y(x)$ denote the calibrated class probability for class y. The Dirichlet concentration vector and its total evidence are

$$\alpha_y(x) = 1 + (S(x) - K)q_y(x), \qquad S(x) = K + \mathrm{softplus}(\hat{\rho}_0 + \hat{\rho}_m M(x)),$$

where $M(x)$ is the BCPNN evidence margin, and the corresponding predictive mean is

$$p_{\mathrm{dir},y}(x) = \frac{\alpha_y(x)}{S(x)}.$$

The floor $\alpha_y(x) \geq 1$ is the *structural atom-break* of C3: because every class retains at least one unit of concentration, no single class can receive all predictive mass (Sensoy et al., 2018; Charpentier et al., 2020). Throughout, $U(x) = K/S(x)$ denotes the evidential uncertainty signal: larger $U(x)$ means lower total Dirichlet evidence.

3.1. Mechanism 1: exact log-odds audit ledger (supports L1)

The audit ledger explains a class decision in the model's own log-odds coordinates. For a tile x, predicted class y, and comparison class $c \neq y$, we audit the Dirichlet predictive log-odds gap

$$\Delta^{\mathrm{dir}}_{y:c}(x) = \log \frac{p_{\mathrm{dir},y}(x)}{p_{\mathrm{dir},c}(x)}.$$

and decompose it into four additive layers,

$$\Delta^{\text{dir}}_{y:c}(x) = \underbrace{\frac{\beta_y - \beta_c}{\hat{T}}}_{\text{class bias}} + \underbrace{\frac{1}{\hat{T}} \sum_{h=1}^{H} \text{Contr}^{(h)}_{y:c}(x)}_{\text{hypercolumn evidence}} + \underbrace{R_{y:c}(x)}_{\text{closure residual}} + \underbrace{\Omega^{\text{dir}}_{y:c}(x)}_{\text{Dirichlet closure}},$$

whose first three terms sum to the temperature-scaled logit gap $(\ell_y(x) - \ell_c(x))/\hat{T}$. Here $\beta_y - \beta_c$ is the class-bias gap, $\text{Contr}^{(h)}_{y:c}(x)$ is the contribution of hypercolumn h to the class-pair logit gap, and $R_{y:c}(x)$ is the numerical closure residual. The term

$$\Omega^{\text{dir}}_{y:c}(x) = \log \frac{p_{\text{dir},y}(x)}{p_{\text{dir},c}(x)} - \log \frac{q_y(x)}{q_c(x)}$$

records the correction introduced by the Dirichlet predictive mean. The hypercolumn contributions are computed from the digamma-based Bayesian-Hebbian edge posteriors of C1,

$$w_{ij} = \psi(C_{ij}) - \psi(C_i) - \psi(C_j) + \psi(C),$$

following Lansner (2009) and Sandberg et al. (2002).

This form separates three quantities that are otherwise easy to confuse: the calibrated class probability $q_y(x)$, the Dirichlet predictive mean $p_{\text{dir},y}(x)$, and the BCPNN edge log-odds w_{ij}. The implementation verifies the additive identity pool-wide to 10^{-6} tolerance and reports a per-hypercolumn diagnostic distribution at the stricter 10^{-3} tolerance.

3.2. Mechanism 2: per-hypercolumn simplex perturbation QP (supports L2)

The counterfactual module tests how far the hidden BCPNN representation must move before a target class overtakes the source class. For a tile x, source class y, and target class $c \neq y$, we solve

$$\begin{aligned} \min_{\Delta h} \quad & \tfrac{1}{2}\|\Delta h\|_2^2 \\ \text{s.t.} \quad & \ell_c(h + \Delta h) - \ell_y(h + \Delta h) \geq \delta, \\ & h_{(j)} + \Delta h_{(j)} \in \Delta^{M-1}, \quad j = 1, \ldots, H. \end{aligned}$$

The simplex constraints preserve the hypercolumn structure of the BCPNN representation: within each hypercolumn, the perturbed minicolumn activations remain non-negative and sum to one. The margin δ sets the required target-class advantage. We use $\delta = 10^{-2}$ by default and $\delta = 0.05$ in the real-tile counterfactual sweep of §5.2.

Equation (3.2) is a convex quadratic program when the class logits are affine in the hidden representation, and we solve it with OSQP (Stellato et al., 2020), whose operator splitting suits repeated convex QPs with warm starts. The output is a hidden-space perturbation diagnostic: it identifies which BCPNN hypercolumns would need to change to flip the decision, but does not by itself define an actionable input-space intervention.

3.3. Mechanism 3: composite reject head (supports L3)

Because §5.5 shows that conformal set size does not reliably identify erroneous tiles on this backbone, we evaluate a separate selective classifier, or reject head, whose job is to decide whether to accept the predicted class or abstain. For each tile x it uses three signals:

$$U(x), \qquad \max_k p_{\mathrm{dir},k}(x), \qquad |\mathcal{C}_\alpha(x)|.$$

The first is evidential uncertainty, the second is the maximum Dirichlet predictive probability, and the third is the randomized-APS prediction-set size. The composite rule accepts a prediction iff

$$U(x) \le \tau_U \quad \wedge \quad \max_k p_{\mathrm{dir},k}(x) \ge \tau_p \quad \wedge \quad |\mathcal{C}_\alpha(x)| \le \tau_C.$$

The thresholds (τ_U, τ_p, τ_C) are tuned jointly on $D_{\text{phys-stage1}}$ by Pareto-grid search. The search uses three operating quantities: a target selective accuracy $a^\star$, a minimum accepted-coverage target $\kappa^\star$, and a selective-risk ceiling $\alpha^\star$, with $a^\star$ and $\alpha^\star$ related as in §2.4.

For a selective rule $g(x) \in \{0,1\}$, where $g(x) = 1$ means "accept", accepted coverage and selective accuracy are

$$\mathrm{cov}(g) = \frac{1}{n}\sum_{i=1}^{n} g(x_i), \qquad \mathrm{acc}(g) = \frac{\sum_{i=1}^{n} g(x_i)\mathbf{1}\{\hat{y}_i = y_i\}}{\sum_{i=1}^{n} g(x_i)}.$$

We report SAC@$a^\star$ as the accepted coverage achieved by a rule satisfying $\mathrm{acc}(g) \ge a^\star$.

We evaluate the selected rule with three selective-risk bounds. *Beta-UCB* is a one-sided Clopper–Pearson upper confidence bound on the empirical selective risk, evaluated on the held-out $D_{\text{phys-stage2}}$ split. *SCRC-I* and *SCRC-T* are the selective conformal risk control bounds of Xu et al. (2025). SCRC-I is the calibration-only variant with PAC-style control, while SCRC-T is the transductive variant that preserves exchangeability by recomputing the threshold with the test sample included. This gives exact finite-sample marginal validity at the cost of per-test recomputation.

3.4. Mechanism 4: ACI shield streaming evaluation (supports L4)

At evaluation time, we also test an adaptive conformal inference (ACI) shield (Gibbs and Candès, 2021). ACI updates the conformal miscoverage level over a stream according to

$$\alpha_{t+1} = \alpha_t + \gamma\left(\alpha^\star - \mathbf{1}\{y_t \notin \mathcal{C}_{\alpha_t}(x_t)\}\right),$$

where $\alpha^\star$ is the target miscoverage level and $1 - \alpha^\star$ is the nominal coverage. The update increases α_t after covered examples and decreases it after miscovered examples, thereby adapting the set size over time.

This experiment is a labeled-streaming evaluation. The recursion uses the observed label y_t at each step, so it is not yet a deployed runtime gate for settings where labels are unavailable or delayed. The shield-only step runs at p50 latencies of 35–67 μs on $\mathcal{D}_{\text{phys}}$ (§5.4).

3.4.1. TIE-modulated nonconformity score

The Dirichlet head of C3 already supplies the main structural atom-break: because $\alpha_y(x) \geq 1$,

$$p_{\mathrm{dir},y}(x) \leq 1 - \frac{K-1}{S(x)},$$

so no single class can receive all predictive mass unless $S(x)$ is infinite. We additionally test whether the same evidential signal can be used at the level of the score itself. Writing

$$s_{\mathrm{APS}}(x,y) = \sum_{k:\, p_{\mathrm{dir},k}(x) \geq p_{\mathrm{dir},y}(x)} p_{\mathrm{dir},k}(x)$$

for the cumulative APS-style nonconformity score computed from the Dirichlet predictive mean, we define

$$s_{\mathrm{TIE}}(x,y;\lambda_U) = s_{\mathrm{APS}}(x,y) + \lambda_U U(x), \qquad \lambda_U \geq 0.$$

The offset is constant across labels for a fixed tile x, so it does not change the within-tile label ordering. Its effect is instead to make the conformal inclusion threshold input-dependent, since $s_{\mathrm{TIE}}(x,y;\lambda_U) \leq \hat{\tau}$ holds exactly when $s_{\mathrm{APS}}(x,y) \leq \hat{\tau} - \lambda_U U(x)$; for $\lambda_U > 0$, high-uncertainty tiles therefore face a stricter effective threshold. We treat s_{TIE} as a score-offset ablation for mean-set-size control, not as a mechanism that guarantees larger sets on uncertain tiles.

The scalar λ_U is fitted on calibration data only, by a sweep subject to a finite-sample coverage constraint, and is frozen before evaluation; the protocol and the resulting sweep are given in Appendix B. On this backbone the Dirichlet floor already does most of the work: $\lambda_U^\star = 0$ often produces informative sets, and the additional mean-set-size reduction for $\lambda_U^\star > 0$ is small ($\sim 0.25\%$ on $\mathcal{D}_{\mathrm{phys}}$, within bootstrap variability). We therefore interpret s_{TIE} as a secondary score ablation, not as the main source of the paper's validity or triage results.

4. Theoretical analysis

This section separates the formal risk-control statements used by the reject head from the atom-specific analysis of the ACI shield. The selective-risk bounds used in §5.3 are stated in Appendix A: a Clopper–Pearson holdout-PAC *Beta-UCB* bound at confidence $1-\delta$ on $D_{\text{phys-stage2}}$; the calibration-only DKW-corrected *SCRC-I* bound at confidence $1-\delta$; and the transductive *SCRC-T* bound, which obtains exact finite-sample marginal validity through per-test recomputation (Xu et al., 2025, Prop. 4, Algo. 2, Thm. 2). We do not re-prove those bounds here. Instead, we analyze why the ACI shield of (3.4) cannot track a nominal target when the nonconformity score distribution contains a large atom at its maximum value.

4.1. No-fixed-point regime of ACI on atom-bearing scores

The intuition is short. ACI nudges α_t up after every covered example and down after every miscovered one, and should settle where those pressures balance. With an atom at the top of the score range that balance point does not exist. While α_t is below the atom mass the threshold is pinned at $s = 1$, nothing is ever miscovered, and the recursion can only climb; once α_t rises past the atom the threshold drops into the continuous part, miscoverage appears abruptly, and the recursion is pushed back down. The target $\alpha^\star$ plays no part in where this happens, so the recursion settles at the atom boundary. The lemma states this as a drift calculation.

Lemma 1 (No fixed-point regime under atomic scores) *Let $\{S_t\}$ be an i.i.d. stream from a stationary distribution with CDF F on $[0, 1]$. Assume that F has an atom of mass $m \in (0, 1)$ at $s = 1$, and that the remaining part*

$$F_\circ(s) = F(s) - m\mathbb{1}[s \geq 1]$$

is continuous and strictly increasing on $[0, 1)$, with total mass $1 - m$. Let the conformal set use the non-strict convention

$$\mathcal{C}_\tau(x) = \{y : s(x, y) \leq \tau\},$$

so that the per-step miscoverage probability is $\Pr[S_t > \tau]$. Define the right-continuous quantile

$$Q_F(u) = \inf\{s : F(s) \geq u\}.$$

For step size $\gamma \in (0, 1]$, target $\alpha^\star \in (0, m)$, and any initialization $\alpha_0 \in [0, 1)$, the expected one-step drift of the ACI recursion

$$\alpha_{t+1} = \alpha_t + \gamma\,(\alpha^\star - \mathbb{1}\{y_t \notin \mathcal{C}_{\alpha_t}(x_t)\})$$

satisfies, in the long-history limit $\hat{q} \to Q_F$:

(i) If $\alpha_t \in [0, m)$, then $Q_F(1 - \alpha_t) = 1$. Therefore

$$\Pr[S_t > Q_F(1 - \alpha_t)] = \Pr[S_t > 1] = 0,$$

and

$$\mathbb{E}[\alpha_{t+1} - \alpha_t \mid \alpha_t] = \gamma\alpha^\star > 0.$$

(ii) If $\alpha_t \in (m, 1)$, then $Q_F(1 - \alpha_t) = F_\circ^{-1}(1 - \alpha_t) \in [0, 1)$. Since the non-atomic part is continuous at this quantile,

$$\Pr[S_t > Q_F(1 - \alpha_t)] = \alpha_t,$$

and

$$\mathbb{E}[\alpha_{t+1} - \alpha_t \mid \alpha_t] = \gamma(\alpha^\star - \alpha_t) < 0.$$

The expected drift is positive below the atom mass and negative above it, with a discontinuity at $\alpha_t = m$. It does not vanish at the operator's nominal target $\alpha^\star$. Consequently, when $\alpha^\star < m$, ACI on atom-bearing scores does not converge to the requested miscoverage target $\alpha^\star$; instead, the recursion is driven toward the atom boundary.

The proof is a direct one-step drift calculation and is given in Appendix A.

Interpretation. Lemma 1 does not provide a closed-form value for the empirical limit α_∞. It identifies the structural reason why target tracking fails: when the conformal quantile falls inside the atom at $s = 1$, the non-strict set convention produces no miscoverage, so the ACI update increases α_t; once the quantile moves below the atom, the empirical miscoverage jumps. The observed trajectory therefore depends on the empirical quantile estimator, tie handling, finite-history effects, and step size γ.

4.2. Numerical verification

We verify the lemma on the BCPNN nonconformity-score distribution from the canonical ledger. The empirical atom mass at $s = 1$, measured as $|s - 1| \leq 10^{-7}$, is $m \approx 0.87$, so each operator-relevant target $\alpha^\star \in \{0.05, 0.10, 0.20\}$ satisfies $\alpha^\star < m$ and lies in the no-fixed-point regime of Lemma 1.

Empirically the recursion tracks none of them. As Fig. 5 shows, across the initializations $\alpha_0 \in \{0.05, 0.10, 0.20\}$ and the four k_{thresh} settings of §5.4, it settles near $\alpha_\infty \approx 0.81$ with residual variation below 0.02. The gap between 0.81 and the canonical-resolution atom mass $m \approx 0.87$ is consistent with finite-history quantile slack and with the coarser effective atom resolution the empirical quantile estimator actually sees: for an operating coarseness ϵ, the effective atom mass $\bar{m}(\epsilon) = \Pr[S \geq 1 - \epsilon]$ can be smaller than the 10^{-7}-resolution estimate used to report m. A precise characterization of α_∞ as a function of $(F, \gamma, t, \hat{q})$ is left to future work.

5. Empirical study

The empirical study reports five lead claims, denoted L1–L5. These claims refine the four contributions stated in the introduction: L1 and L2 correspond to the interpretability artifacts, L3 to the selective-risk alternative, L4 to the streaming ACI shield, and L5 to the atom-breaking study. Each claim uses the split protocol of §2.4. The order is deliberate: L1–L4 characterise what each individual mechanism does and does not deliver, and L5 (§5.5) then gives the negative result that justifies making the reject head, rather than the conformal set, the operating layer.

L1 computes the audit ledger in process on the entire post-class-0 $\mathcal{D}_{\text{phys}}$ pool. L2 runs a per-tile counterfactual sweep over 100 seeded $\mathcal{D}_{\text{phys}}$ hidden-activation rows. L3 uses the 50/25/25 physical-photo split for reject-head refit, holdout calibration, and evaluation. L4 is a labeled-streaming ACI replay with warm-up 1000 patches, $\gamma = 0.01$, initializations $\alpha_0 \in \{0.05, 0.10, 0.20\}$, and $k_{\text{thresh}} \in \{1, 2, 3, 4\}$. L5 compares five conformal variants on the same canonical per-tile signal ledger.

5.1. L1: auditable explanations

The pool-wide max residual on the post-class-0 $\mathcal{D}_{\text{phys}}$ pool is 1.71×10^{-7}, well below the 10^{-6} tolerance. The per-hypercolumn ledger on 50 patches crossed with all 20 ordered class pairs has median residual 0.025 nats, p95 0.103, max 0.311 — below the 0.5-nat engineering gate of (Cruz Ulloa et al., 2025, §7.9.9) but not the strict 10^{-3} contract, so we report it as a diagnostic distribution rather than a binary identity. Its upper-tail tiles, whose hidden

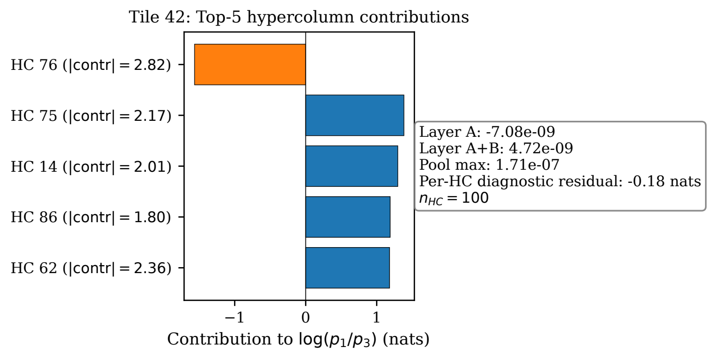


Figure 3: Per-prediction explanation card on a WTA-saturated $\mathcal{D}_{\text{phys}}$ tile (tile 42, $y = 1$, $c = 3$). The top-5 hypercolumn contributions to $\log(p_y/p_c)$ sum, modulo a class-bias term, to the Layer C+D logit gap; the inset reports the residual terms used for the audit check. The per-hypercolumn diagnostic residual -0.18 nats reflects the WTA-saturation regime that also drives the L2 counterfactual finding.

activations fall in the WTA-saturated regime, are precisely those driving the L2 finding of §5.2.

5.2. L2: counterfactual validity

On synthetic Dirichlet(0.5) activations the per-hypercolumn simplex QP of (3.2) achieves 1.00 argmax-flip validity. On a 100-tile seeded sweep of *real* hidden activations from $\mathcal{D}_{\text{phys}}$ using real weights, margin $\delta = 0.05$, and the 5 s OSQP budget, validity drops to 0.83 (Wilson 95% CI $[0.745, 0.891]$, $n = 100$) with p95 runtime 2.73 s and mean ℓ_2 cost 0.144. The solver-status histogram is part of the result, not an implementation detail: 64 solves are optimal, 19 optimal-inaccurate, and 17 hit the user-limit budget (Appendix D). The 17% time-limit failures are the clearest non-recourse outcomes under the current budget; the 19% optimal-inaccurate cases are accepted only after post-validation and should be read as lower-confidence counterfactuals. Both modes concentrate on WTA-saturated tiles where most of one hypercolumn's mass is on a single minicolumn (Tully, 2016). The corresponding development check was marked expected-fail at the 0.95 gate; the 0.83 value is this section's empirical finding, not a pass/fail regression target. Figure 6 (Appendix D) shows a feasible flip and a budget-exhausted saturated case side by side.

5.3. L3: bounded-risk acceptance

We tune (τ_U, τ_p, τ_C) using the SAR reporting gate $a^\star = 0.70$, minimum accepted coverage $\kappa^\star = 0.30$, selective-risk budget $\alpha^\star = 0.32$, and confidence $1 - \delta = 0.90$. The accuracy gate and the risk budget are intentionally reported as separate quantities. The bound is evaluated at $\alpha^\star = 0.32$, corresponding to nominal selective accuracy $1 - \alpha^\star = 0.68$, while SAC@0.70 reports accepted coverage at the stricter 70% empirical selective-accuracy gate.

At this operating point, the three bounds are informative in the sense that the empirical selective risk of the chosen rule lies below the reported upper bound. In an alternate

Table 3: Bound×head-refit ablation on the physical-photo splits at canonical $\alpha^\star = 0.32$ ($n_{\text{eval}} = 5707$, seed 20260425, $\delta = 0.10$). Rows vary both the bound and the head-refit protocol. The no-refit Beta-UCB row uses the $\mathcal{D}_{\text{cal-sim}}$-fit Dirichlet head and independent stage-2 holdout counts; the SCRC-I/T rows use the stage-1-refitted head required by the SCRC protocol and eval-pool selection counts. Rows across the horizontal rule are therefore *not* a pure same-head bound-method comparison. SCRC-T attains exact finite-sample marginal validity at $\alpha^\star$ in the per-test exchangeability sense of Xu et al. (2025); SCRC-I reports the corresponding DKW-corrected calibration-only approximation.

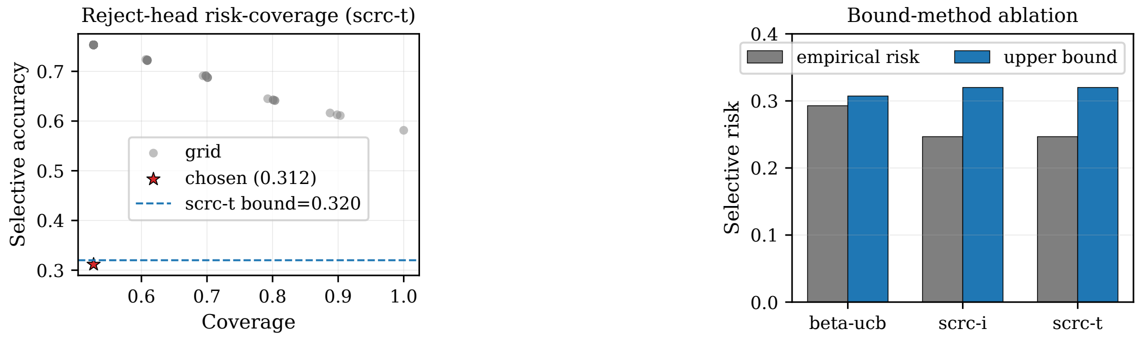
Bound	Head	Src.	n_{acc}	n_{err}	Emp. risk	Bound	Bound type	Pass?
Beta-UCB	no refit	stage-2	1745	511	0.293	0.307	Clopper–Pearson	✓
SCRC-I	stage-1 refit	eval	3008	742	0.247	0.320	approx. DKW	✓
SCRC-T	stage-1 refit	eval	3008	742	0.247	0.320	**exact**	✓

$\alpha^\star = 0.15$ run, stage-1-refitted SCRC-I and SCRC-T produce zero held-out errors and trivially satisfy the target, whereas Beta-UCB without refit fails the budget at 0.307. To avoid mixing operating points, §5.3 and Table 3 use the canonical $\alpha^\star = 0.32$ configuration throughout.

Read along the head-refit axis, the stage-1-refit rows are the canonical SCRC configuration: SCRC-T attains *exact* finite-sample marginal validity at 0.32 with empirical risk 0.247 (Wilson 95% CI $[0.232, 0.262]$, 742 errors among 3008 accepted samples), and SCRC-I attains the same numerical bound under a DKW-corrected approximation, with the same empirical risk. Read along the bound-method axis, the no-refit Beta-UCB row anchors the small-stage-2 Clopper–Pearson slack ($1/\sqrt{n_{\text{stage-2}}}$): it accepts fewer samples (1745) and gives a 0.307 bound with empirical risk 0.293 (Wilson 95% CI $[0.272, 0.315]$) on the stage-2 holdout, while the same saved rule accepts 3577 samples on the eval pool at selective risk 0.284. We do not claim a direct method-vs-method comparison across the horizontal rule because the head differs; a same-head Beta-UCB row would require an additional stage-1-refit run that we leave to a follow-up. The SCRC-T "marginal" guarantee is in the per-test-point exchangeability sense of Xu et al. (2025) Theorem 2, under which the 0.073 margin reflects favorable finite-sample variance rather than bound looseness (Appendix A.2). The stage-1-refit rule degenerates the U threshold to its upper bound but accepts $\sim 1.7\times$ as many samples; the no-refit Beta-UCB rule uses all three composite signals at the cost of tighter coverage. We keep the canonical $\alpha^\star = 0.32$ run as the sole reported operating point. Figure 4 plots the rule grid in coverage–selective accuracy space (a) and contrasts the three bounds on the same per-tile signals (b).

5.4. L4: streaming-shield latency and structural-quantile convergence

The ACI shield is run as a labeled streaming replay on the full post-class-0 $\mathcal{D}_{\text{phys}}$ pool ($n = 11\,413$), not only on $D_{\text{phys-eval}}$, with warm-up 1000 patches, $\gamma = 0.01$, initializations $\alpha_0 \in \{0.05, 0.10, 0.20\}$, and the four $k_{\text{thresh}} \in \{1, 2, 3, 4\}$ settings used for the latency grid. Across this grid the reference Python implementation attains p50 latency 35–67 μs per



(a) Risk–coverage with chosen rule.

(b) Bound-method ablation.

Figure 4: Composite reject head. (a) Selective accuracy versus coverage for an 11×11 grid of (τ_U, τ_p) rules; the chosen point is annotated. (b) Empirical selective risk and upper bound across Beta-UCB, SCRC-I, SCRC-T at the same $(\delta, \alpha^\star)$ targets from the canonical JSON ledgers.

step and p99 latency 124–282 μs, negligible relative to a 10 Hz perception-loop budget. The recursion converges in mean to $\alpha_\infty \approx 0.81$ within $\sim$1000 steps and remains there (Figure 5). Per Lemma 1, α_∞ is *not* the operator's nominal target $\alpha^\star$ whenever $\alpha^\star$ lies below the atom mass: the expected drift never vanishes inside $[0, 1)$ and the recursion settles into the atom-tracking regime of §4.2.

5.5. Atom-breaking does not rescue size-based triage

We close the empirical study with the negative result that motivates the composite reject head of §5.3. Raw split-CP and randomized APS serve as collapse baselines, both returning the constant $K = 5$ set on the canonical signals ledger; ConfTS, the structural Dirichlet floor, and ACI produce smaller sets with marginal coverage close to target, yet $|\mathcal{C}(x)|$ still fails to rank erroneous tiles above correct ones. APS is built to produce marginally valid, adaptive sets, not a calibrated per-tile error score, and the gap between those two properties is what this section measures.

Table 4 compares five conformal variants on the same per-tile signals from the physical-photo evaluation pool $D_{\text{phys-eval}}$ ($n_{\text{eval}} = 5707$, seed 20260425). Each row reports marginal coverage, mean set size, and, when defined on the canonical ledger, the AUROC of $|\mathcal{C}(x)|$ as a binary triage score for the event $\mathbf{1}\{\hat{y}(x) \neq y\}$. This AUROC is the rank-discrimination quantity tested by the size-based shielding hypothesis: above 0.5, larger sets tend to rank erroneous tiles as harder; near 0.5, there is no useful ranking signal; below 0.5, the direction is reversed.

The failure is not confined to $\alpha = 0.10$. For TIE-Dirichlet split CP, AUROC$[|\mathcal{C}|]$ is 0.571, 0.322, and 0.500 at $\alpha \in \{0.05, 0.10, 0.20\}$ (Appendix B), so the sign changes with α: larger sets weakly track errors at 0.05, anti-track them at 0.10 (AUROC$[-|\mathcal{C}|] = 0.678$), and carry essentially no signal at 0.20. Set size is therefore not a stable monotone triage score here. A fixed-direction $-|\mathcal{C}|$ score would gain only after labeled, per-α recalibration, which defeats the intended use of set size as a fixed, monotone, label-free signal.

Table 4: Atom-break mechanisms compared on BCPNN $\mathcal{D}_{\text{phys}}$ at $\alpha = 0.10$ (target coverage 0.90, $n_{\text{eval}} = 5707$). Mean set size $\bar{s}$ decreases when the atom is avoided or structurally broken. Coverage carries a Wilson 95% CI; the size-AUROC carries the Hanley–McNeil normal-approximation 95% CI at the eval-pool case/control split ($n_{\text{pos}} = 2389$ errors, $n_{\text{neg}} = 3318$ correct), computed for ACI after the 1000-sample warm-up ($n_{\text{pos}} = 2091$, $n_{\text{neg}} = 2616$). The random-ranking H_0 band is $[0.485, 0.515]$ on the full eval pool and $[0.483, 0.517]$ after warm-up. The Dirichlet floor $\alpha_y \geq 1$ gives compact marginally valid sets, yet its set size ranks errors in the *wrong* direction at this operating point: the negative result is not explained by lack of marginal calibration alone.

Variant	Mechanism	Coverage	$\bar{s}$	AUROC[$\lvert\mathcal{C}\rvert$]
Raw split-CP APS	atom-limited baseline	1.000	5.00	n/a†
Randomized APS (Romano et al., 2020)	tie noise; still atom-limited	1.000	5.00	n/a†
ConfTS ($\hat{T}$=12.01) (Xi et al., 2025)	logit tempering	0.930	4.03	—‡
TIE split-CP (ours)	Dirichlet floor	0.909 [.902, .917]	3.54	0.323 [.304, .341]
TIE rand. APS (ours)	floor + APS noise	0.900 [.892, .907]	3.68	0.441 [.421, .461]
ACI (γ=0.01) (Gibbs and Candès, 2021)	online quantile update	0.899 [.890, .907]	3.50	0.498 [.476, .519]

† When all set sizes equal the constant $K = 5$, AUROC[$\lvert\mathcal{C}\rvert$] is undefined because there are no non-tied size ranks. We mark these rows "n/a" rather than reporting an arbitrary tie-broken score. ‡ ConfTS coverage and mean set size reproduce Cruz Ulloa et al. (2025) Table 4. A canonical-ledger size-AUROC is not reported because the canonical ledger omits the raw softmax logits required by the ConfTS pipeline; sister-experiment evidence is discussed in Appendix C.

Mechanism by mechanism. Raw split-CP and randomized APS collapse because $\hat{\tau}$ falls inside the atom (§2). ConfTS avoids part of the collapse by tempering the logits at $\hat{T} = 12.01$ (Xi et al., 2025); its size-AUROC evidence is in Appendix C. The Dirichlet head's atom-break is structural, following from the $\alpha_y \geq 1$ floor rather than a fitted threshold — yet the TIE split-CP and TIE randomized-APS AUROCs (0.323 and 0.441) show that larger sets are not reliable error indicators here, and in the split-CP case are anti-correlated with errors. ACI moves the adaptation into an online quantile update on the Dirichlet-head probabilities; after warm-up AUROC[$\lvert\mathcal{C}\rvert$] $= 0.498$, indistinguishable from the random-ranking band $[0.483, 0.517]$. This is separate from the raw atom-bearing ACI analysis of §4, where the recursion tracks the atom boundary rather than the operator's target $\alpha^\star$.

Why $\lvert\mathcal{C}\rvert$ is a poor triage signal here. The failure is specific to set size: on the same pool the evidential signals $U(x) = K/S(x)$ and $\max_k p_{\text{dir},k}(x)$ each reach AUROC ≈ 0.70, motivating the composite reject head of §5.3. In the tuned rule $\tau_C = K$: the set-size constraint is inactive and acceptance is driven by $U(x)$ and $\max_k p_{\text{dir},k}(x)$.

TIE modulation does not rescue triage. The sweep of §3.4.1 selects $\lambda_U^\star = 0.10$ at $\alpha = 0.10$, giving validation coverage 0.912 and mean set size $\bar{s} = 3.521$ against $\bar{s} = 3.530$ at $\lambda_U = 0$ — a relative reduction of about 0.25%, well within sampling variability at $n_{\text{eval}} = 5707$. We therefore treat s_{TIE} as a calibration-side score-offset ablation that may matter on backbones with heavier residual atoms, not as a triage instrument here.

6. Discussion: FAT-relevant evidence

We set out to tell an operator when to trust a tile, and arrived at a system that answers with an abstention rule and two audit artifacts rather than a prediction set. What is that evidence good for, and how far does it reach? Whether a deployed SAR perception system satisfies a regulatory transparency, accountability, or documentation requirement depends on the full system context: the deployer's use case, human-oversight procedure, risk-management file, and post-market monitoring process. Our contribution is narrower — that a BCPNN-based backbone can be equipped with auditable uncertainty, selective-risk, and explanation artifacts that are useful inputs to such documentation.

The reported evidence is relevant to the fairness, accountability, and transparency (FAT) disclosure axes in three ways. First, SAC@$a^\star$ reported with its selective-risk bound (§5.3) gives a deployer-facing operating metric: how much of the evaluation pool is accepted at a target selective accuracy, and which risk-control statement supports it. This aligns with the transparency logic of Article 13 of the EU AI Act, which requires high-risk systems to be accompanied by clear information enabling deployers to interpret the system's output and use it appropriately (European Parliament and Council, 2024). Second, the per-hypercolumn log-odds ledger (§5.1) and the hidden-space counterfactual diagnostic (§5.2) provide candidate explanation artifacts, identifying which internal evidence terms support a prediction and how far the hidden representation would have to move to change it. These are relevant to the explanation practice reflected in Article 86, while not by themselves determining whether Article 86 applies to a given SAR deployment. Third, the evidential uncertainty $U(x)$, conformal prediction set, accepted-coverage metric, and selective-risk bounds make explicit the uncertainty and evaluation assumptions behind the reported performance — the direction taken by recent NIST guidance, which asks that benchmark-style claims state the evaluation target, assumptions, sources of variation, and uncertainty rather than point estimates alone (Keller et al., 2026).

The main technical lesson is negative but operationally important. Breaking the atom at $s = 1$ can recover compact, marginally valid conformal sets, but does not make conformal set size a stable per-tile triage signal on this backbone, so the operating layer is a selective reject head evaluated under explicit risk-control bounds rather than a size-only conformal shield. The distinction matters for deployment: coverage validity, risk-controlled abstention, and explanation diagnostics answer different questions and should not be collapsed into a single "trustworthiness" score.

6.1. Portability: when to expect an atom elsewhere

Our empirical claims concern one backbone and one corpus, and we do not generalise them. The *mechanism*, however, contains nothing specific to BCPNN, and stating what it needs lets others check their own models cheaply. The collapse of §2.2 requires two ingredients: a nonconformity score bounded above that attains its bound whenever the classifier is confident — true of APS-style cumulative scores by construction — and a classifier whose predictive distribution sits close enough to a simplex vertex that a large fraction of calibration points reach that bound. Given both, split conformal prediction returns the full label space for every $\alpha < m$, and by Lemma 1 an online ACI recursion cannot track such a target either. Neither ingredient mentions Hebbian learning, hypercolumns, or terrain: a near-vertex predictive posterior is the only property the argument needs. We would therefore expect the same failure wherever training or recalibration drives the predictive distribution towards a vertex, but we have not tested that expectation and offer it as a place to look, not a result.

A one-line diagnostic. The question is cheap to settle, and is best asked before a conformal layer is deployed. On the calibration split, report the effective atom mass at the resolution ϵ at which the quantile estimator actually operates, $\bar{m}(\epsilon) = \Pr[s \geq 1 - \epsilon]$ (§4.2). If $\bar{m}(\epsilon) > \alpha$ for the target miscoverage α, the set will contain every class attaining the bound and the reported coverage will be uninformative rather than wrong. Printing $\bar{m}$ beside coverage and mean set size costs one line and would have exposed this failure immediately. A second check follows from §5.5: if the set is meant for triage rather than coverage, the quantity to validate is the AUROC of $|\mathcal{C}(x)|$ against the error indicator — not coverage, as the two can disagree completely.

Limitations. The evaluation is limited to one SAR sim-to-real corpus (Cruz Ulloa et al., 2025) and one classifier family, a BCPNN backbone with the TIE-BCPNN extensions; the conjecture of §6.1 that atom-induced collapse occurs elsewhere is not tested here. The formal risk-control claims inherit the exchangeability assumptions of their respective split protocols, and sustained distribution shift would require explicit shift-aware correction, such as weighted conformal prediction, which this submission does not deploy. The reject head is tuned offline on a fixed split, so the reported SAC operating point should be read as a held-out evaluation result rather than a closed-loop field guarantee. The counterfactual QP is ℓ_2-regularized and operates in hidden space; an ℓ_1 sparsity-promoting variant may improve interpretability but could reduce feasibility, and we do not ablate that trade-off. Finally, the ACI shield is evaluated as a labeled streaming replay, not a label-free runtime gate; on-board execution, delayed-label operation, quadruped retrofit, and retraining under the full TIE-BCPNN pipeline are left to future work.

7. Conclusion

We began with an operator who needs to know which tiles to trust, and a conformal predictor that answered every such question with "all five classes". The cause was an atom of mass approximately 0.87 at $s = 1$ in the APS nonconformity distribution, collapsing split conformal prediction to the trivial K-class set across $\alpha \in [0.05, 0.20]$. Breaking the atom can recover compact, marginally valid sets — but, and this is the paper's central finding, it does not make set size a reliable per-tile triage signal.

The operational alternative is a composite reject head under explicit selective-risk bounds: in the canonical configuration SCRC-T attains exact finite-sample marginal validity at $\alpha^\star = 0.32$, with empirical selective risk 0.247 on the accepted subset. The same evaluation supplies an exact four-layer log-odds audit ledger and a per-hypercolumn hidden-space counterfactual diagnostic, while the labeled-streaming ACI evaluation shows the adaptive recursion to be stable but tracking the atom boundary rather than the operator's target. The conclusion is therefore not that conformal set size can shield a near-deterministic Bayesian classifier; on this case study it cannot. What works is the combination, bounded by the split protocol, the single SAR corpus, and the BCPNN backbone evaluated here.

Acknowledgments

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Appendix A. Proofs and regularity conditions

A.1. Beta-UCB Clopper–Pearson derivation

Let $g(x) \in \{0, 1\}$ be the selected acceptance rule, fixed before evaluating the held-out $D_{\text{phys-stage2}}$ split. Among the accepted samples, define

$$n_{\text{acc}} = \sum_i g(x_i), \qquad n_{\text{err}} = \sum_i g(x_i)\mathbf{1}\{\hat{y}_i \neq y_i\}.$$

The selective risk of the fixed rule is the conditional error probability

$$R(g) = \Pr[\hat{y} \neq y \mid g(x) = 1].$$

Conditional on $n_{\text{acc}} > 0$, the accepted-error count is modeled as a binomial count with parameter $R(g)$. The one-sided Clopper–Pearson upper confidence limit at confidence $1 - \delta$ is

$$U_\delta = \text{Beta}^{-1}\left(1 - \delta;\, n_{\text{err}} + 1,\, n_{\text{acc}} - n_{\text{err}}\right),$$

with the usual boundary convention $U_\delta = 1$ when $n_{\text{err}} = n_{\text{acc}}$. We report this upper limit as the Beta-UCB bound. Its validity is conditional on the rule being fixed before the independent holdout evaluation and on the accepted stage-2 examples being exchangeable under the physical-photo split.

A.2. SCRC-I and SCRC-T

We use the SCRC-I and SCRC-T procedures of Xu et al. (2025) as selective conformal-risk bounds. SCRC-I is the calibration-only variant: it selects thresholds using the calibration split and adds a DKW-style correction for empirical-CDF error. SCRC-T is the transductive variant: for each test point, the threshold is recomputed with that test point included in the calibration construction, which yields exact finite-sample marginal validity under the exchangeability assumption stated in Xu et al. (2025, Thm. 2). We do not re-prove these results; this appendix records the assumptions used in our split protocol and the mapping between their notation and our reject-head thresholds.

The guarantee of Xu et al. (2025, Thm. 2) is $\mathbb{E}[\text{risk}] \le \alpha^\star$, an expectation over the test draw rather than a pool-wise statement. Finite-sample pool-wide empirical risk on a single evaluation pool may therefore fall on either side of the reported bound without violating the guarantee. The 0.073 margin observed in Table 3 should be read in that light: it reflects favorable finite-sample variance on this particular pool, not looseness of the bound.

A.3. Proof of Lemma 1

The ACI recursion has conditional drift

$$\mathbb{E}[\alpha_{t+1} - \alpha_t \mid \mathcal{F}_t] = \gamma\left(\alpha^\star - \Pr[S_t > \hat{q}_t(\alpha_t) \mid \mathcal{F}_t]\right).$$

In the long-history limit, $\hat{q}_t(\alpha_t) \to Q_F(1 - \alpha_t)$. If $\alpha_t < m$, then $Q_F(1 - \alpha_t) = 1$. Under the non-strict set convention $s \le \tau$, the miscoverage probability is $\Pr[S_t > 1] = 0$, so the drift is $+\gamma\alpha^\star$. If $\alpha_t > m$, the quantile lies in the continuous part of the distribution and

$$\Pr[S_t > Q_F(1 - \alpha_t)] = \alpha_t,$$

so the drift is $\gamma(\alpha^\star - \alpha_t) < 0$ whenever $\alpha^\star < m < \alpha_t$. At the boundary $\alpha_t = m$, the right-continuous quantile convention and empirical tie handling determine the realised update; this is why Lemma 1 states a no-fixed-point regime rather than a closed-form equilibrium. The nominal target $\alpha^\star < m$ is not a fixed point.

A.4. ACI trajectory diagnostic

Figure 5 shows the empirical companion to Lemma 1: the realised α_t trajectories of the labeled streaming replay of §5.4, for the three initializations used there.

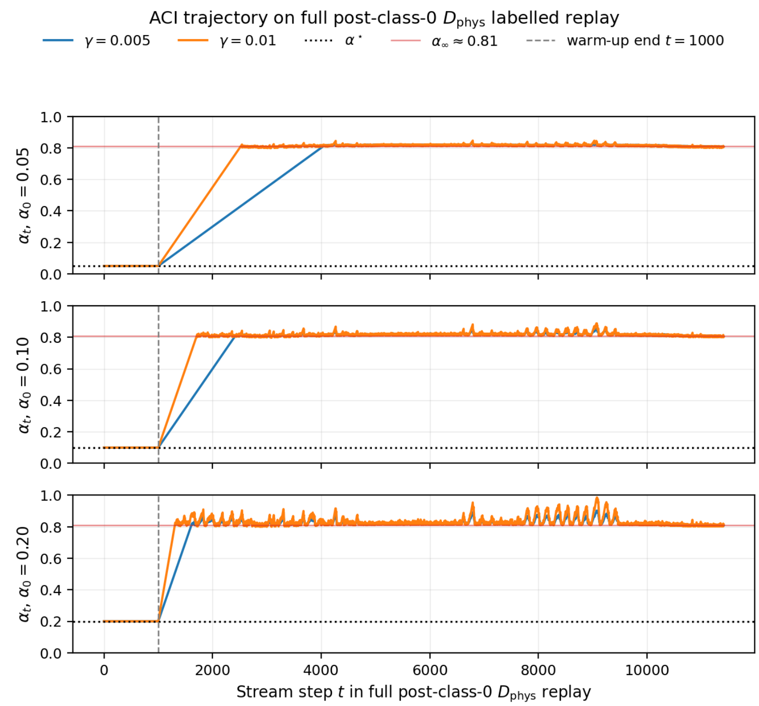

Figure 5: ACI miscoverage parameter α_t on the full post-class-0 $\mathcal{D}_{\text{phys}}$ labeled replay ($n = 11\,413$, warm-up $= 1000$). The three panels correspond to initializations $\alpha_0 \in \{0.05, 0.10, 0.20\}$. All trajectories settle near the atom-tracking equilibrium $\alpha_\infty \approx 0.81$, showing stability across initializations rather than tracking of the nominal target $\alpha^\star$.

Appendix B. Atom-break comparative study: full grid and λ_U sweep

This appendix records the calibration-side protocol and λ_U sweep used for §3.4.1 and §5.5. The scalar λ_U of (3.4.1) is fitted using calibration data only. We split D_{cal} into D_{fit} and D_{val}, fit $\hat{\tau}(\lambda_U)$ on D_{fit}, and choose $\lambda_U^\star$ to minimize mean set size on D_{val} subject to the finite-sample validation constraint

$$\widehat{\text{cov}} \geq 1 - \alpha - \frac{2}{\sqrt{|D_{\text{val}}|}}.$$

The selected $\lambda_U^\star$ is frozen before evaluation. The sweep at $\alpha = 0.10$ evaluates

$$\lambda_U \in \{0, 0.05, 0.1, 0.2, 0.5, 1.0\}.$$

It selects $\lambda_U^\star = 0.10$, with validation coverage 0.912 and mean set size $\bar{s} = 3.521$, compared with $\bar{s} = 3.530$ at $\lambda_U = 0$. This is a relative reduction of approximately 0.25%, within bootstrap variability on D_{val}. Validation coverage remains above the finite-sample validation floor

$$1 - \alpha - \frac{2}{\sqrt{|D_{\text{val}}|}} = 0.863$$

throughout the sweep. These values support the conclusion that s_{TIE} provides only a small set-size reduction on this backbone.

Appendix C. ConfTS reference comparison

The ConfTS row of Table 4 reports coverage and mean-set-size values reproduced from Cruz Ulloa et al. (2025), Table 4. We do not report a size-AUROC for ConfTS in the main table because this analysis used APS-side scores rather than the raw softmax logits required to rerun the ConfTS pipeline. Sister-experiment runs on the same BCPNN backbone place AUROC[$|\mathcal{C}|$] for ConfTS-tempered split CP near the random-ranking baseline at $\alpha = 0.10$, consistent with the negative size-triage result of §5.5. We therefore treat ConfTS as supporting evidence for compact marginally valid sets, not as a size-based triage measurement for this paper. Persisting raw logits in the conformal-evaluation CLI and rerunning ConfTS on the same split is left to follow-up work.

Appendix D. Counterfactual implementation details

The OSQP configuration uses absolute and relative tolerances 10^{-5}, at most 50 000 iterations, and a 5 s time limit. We warm-start with the previous-tile Δh when the source and target classes match, and otherwise use zero initialization. In the WTA-saturated regime (Tully, 2016), most of a hypercolumn's mass lies on a single minicolumn. The simplex constraints therefore bind near vertex faces, which is consistent with the observed solver stress.

In the 100-tile real-activation sweep, 64 solves are certified optimal, 19 are optimal but flagged inaccurate by OSQP, and 17 reach the configured solver time or iteration budget. We treat the budget-exhausted cases as non-recourse outcomes under the current 5 s solver budget. The flagged inaccurate solves are retained only when post-solve validation confirms the requested argmax flip and margin.

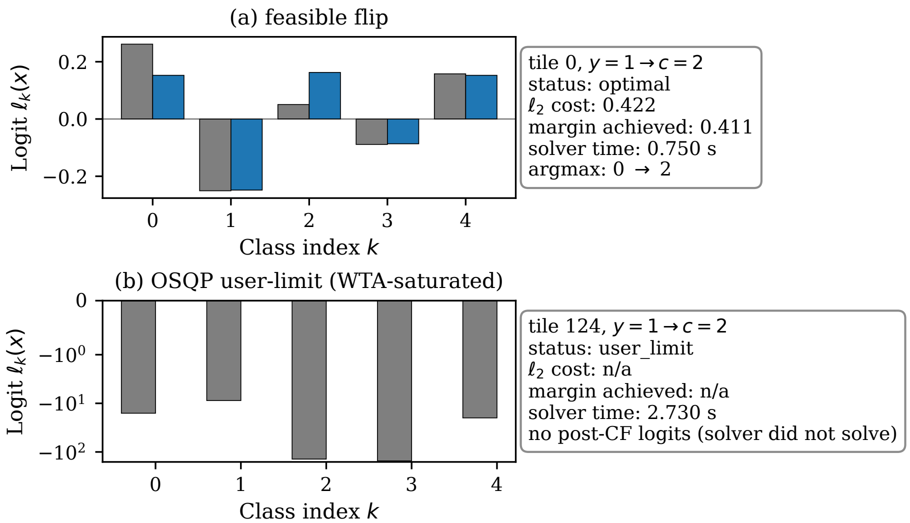

Figure 6: Counterfactual visualization. Gray bars show pre-counterfactual logits and blue bars show post-counterfactual logits. Panel (a): a feasible flip on a non-saturated tile (tile 0), with ℓ_2 cost 0.42, achieved margin 0.41, and argmax shift from class 0 to 2. Panel (b): a WTA-saturated user-limit case (tile 124) where the OSQP solver exhausts its budget before producing a validated flip; the saturated-logit dynamic range is shown on a symmetric log scale, and no post-counterfactual logits are reported.